

WMDR2-core

Package in package 'Model'

WMDR2-core
Version 0.1.11 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-18

WMDR2-core diagram

Class diagram in package 'WMDR2-core'

wmdr2-core

Notes

1. temporalExtent shall be realized as time:interval
2. temporal<Name> elements other than temporalGeometry shall be realized as arrays of dictionary objects for (begin) date, and the values of <Name>, e.g.,
temporalTerritory = [{
"date": "2000-01-01",
"territory": "<ISO1>"
},
{
"date": "2024-07-13",
"territory": "<ISO2>"
}]
3. Following the OGC Features model, geometry = geolocation = coordinates + elevation
4. Schedules

v0.1.4 Introduce instrument class.

v0.1.5 Expand instrument class.

v0.1.6 introduce more elements

v0.1.8

- require and expanded observation.observedDomain (includes domain, domainFeature [0..1], featureName [0..1])
- introduce temporalGeometry.methods

v0.1.9

- introduce deployment.temporalGeometry [0..1], verticalDistanceFromReferenceSurface [0..1]
- rename localReferenceSurface to referenceSurface
- rename observation.observedVariable to observation.observedProperty
- remove xType suffix to manage expectations: many elements are 'Types', i.e., codelist-based

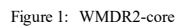
v0.1.10

- rename environment.temporalTopographyBathymetry to topographyBathymetry
- add environment.temporalSurfaceRoughness
- rename instrumentCapability to instrument
- rename instrument.observableVariable to observedProperty, observableGeometry to observedGeometry.

v.0.1.11

- rename observedDomain to domain
- move reporting.officialStatus to deployment.temporalOfficialStatus
- remove inline deployment.temporalObservingSchedule as being redundant to the explicit association.
- update cardinality of association to deployment.temporalObservingSchedule
- model verticalDistanceFromReferenceSurface with "value" and "uom"
- update documentation

WMDR2-core
Version 0.1.11
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Class diagram in package 'WMDR2-core'

Notes

- version 0.1.4: simplify observation, specifically specification of 1-01

WMDR2-core-logical

Version 0.1.4

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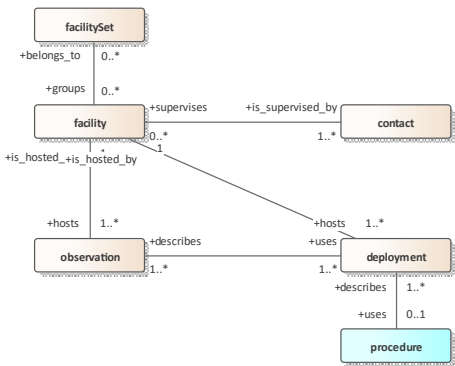


Figure 2: WMDR2-core-logical

contact

Class in package 'WMDR2-core'

WMDR2 core JSON class: contact.

contact

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES	
addresses : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Postal addresses. [Is static True. Containment is Not Specified.]	
emails : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Email contacts. [Is static True. Containment is Not Specified.]	
id : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Optional stable contact identifier. [Is static True. Containment is Not Specified.]	
links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Online resources associated with the contact.	

ATTRIBUTES	
[Is static True. Containment is Not Specified.]	
 name : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Either a full name, or a sequence of name parts. Examples: Dr. Jörg Klausen (this corresponds to FN in vcard) Klausen;Jörg;Dr.; (this corresponds to N in vcard) [Is static True. Containment is Not Specified.]	
 organization : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Example: ORG:Federal Office of Meteorology and Climatology MeteoSwiss [Is static True. Containment is Not Specified.]	
 phones : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Phone contacts. [Is static True. Containment is Not Specified.]	
 roles : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Specific contact roles such as owner, pointOfContact, principalInvestigator. [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
 Association (direction: Unspecified) 9-01 responsibleParty Source: Public supervises (Class) facility Cardinality: [0..*] Target: Public is_supervised_by (Class) contact Cardinality: [1..*]	

deployment

Class in package 'WMDR2-core'

WMDR2 core JSON class: deployment.

deployment

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES
 contacts : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

ATTRIBUTES	
Deployment-level contacts.	[Is static True. Containment is Not Specified.]
 exposure : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Compact exposure classification value.	[Is static True. Containment is Not Specified.]
 id : Private Deployment identifier; source XML id is preserved when available.	[Is static True. Containment is Not Specified.]
 instrument : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) References to reusable instrument ids.	[Is static True. Containment is Not Specified.]
 keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Deployment keywords.	[Is static True. Containment is Not Specified.]
 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Deployment links.	[Is static True. Containment is Not Specified.]
 observingMethod : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) from (new) unified code list of observing methods, based on http://codes.wmo.int/wmdr/ObservingMethodAtmosphere http://codes.wmo.int/wmdr/ObservingMethodTerrestrial http://codes.wmo.int/wmdr/ObservingMethodEarth	[Is static True. Containment is Not Specified.]
 referenceSurface : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Linked to heightAboveReferenceSurface, from http://codes.wmo.int/wmdr/ReferenceSurfaceType	[Is static True. Containment is Not Specified.]
 representativeness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) http://codes.wmo.int/wmdr/Representativeness	[Is static True. Containment is Not Specified.]

ATTRIBUTES

sourceOfObservation : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Compact source-of-observation code value.

[Is static True. Containment is Not Specified.]

temporalInstrumentOperatingStatus : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[{"instrumentOperatingStatus": str, "date": date}, {...}], whereby "instrumentOperatingStatus" values refer to <http://codes.wmo.int/wmdr/InstrumentOperatingStatus>

[Is static True. Containment is Not Specified.]

temporalOfficialStatus : Private
 Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

[{"officialStatus": str, "date": date}, {...}], whereby "officialStatus" values refer to <http://codes.wmo.int/wmdr/ObservationStatus>

[Is static True. Containment is Not Specified.]

temporalSerialNumbers : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[{"serialNumber": str, "date": date}, {...}]

[Is static True. Containment is Not Specified.]

time : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Deployment temporal extent, date-resolution interval or null.

[Is static True. Containment is Not Specified.]

verticalDistanceFromReferenceSurface : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

implement as an array of objects:

[{"value": float, "uom": str}, {...}], whereby "uom" from <http://codes.wmo.int/wmdr/unit>

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public describes (Class) deployment
 Cardinality: [1]

Target: Public reports_with (Class)
 temporalObservingSchedule
 Cardinality: [0..*]

Association (direction: Unspecified) 5-09 instrument

Source: Public is_used_by (Class) deployment

Target: Public uses (Class) instrument

ASSOCIATIONS	
Cardinality: [1..*]	Cardinality: [0..1]
Association (direction: Unspecified) 3-07 temporalGeometry	
Source: Public (Class) deployment Cardinality: [1]	Target: Public (Class) temporalGeometry Cardinality: [0..1]
describes	has_geolocatino
Association (direction: Unspecified)	
Source: Public describes (Class) deployment Cardinality: [1..*]	Target: Public uses (Class) procedure Cardinality: [0..1]
	ObservingProcedure(s)
Association (direction: Unspecified)	
Source: Public is_hosted_by (Class) facility Cardinality: [1]	Target: Public hosts (Class) deployment Cardinality: [1..*]
Association (direction: Unspecified)	
Source: Public describes (Class) observation Cardinality: [1..*]	Target: Public uses (Class) deployment Cardinality: [1..*]

domain

Class in package 'WMDR2-core'

Describes the physical domain, i.e. larger-scale environmental target or ultimate-feature-of-interest, of an observation.

domain

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES
domain : Private The largest scale in WIGOS, cf. http://codes.wmo.int/wmdr/Domain [Is static True. Containment is Not Specified.]
domainFeature : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Describes the type of the larger-scale environmental target or ultimate-feature-of-interest, of an observation. Concept motivated by requirements from hydrology, but may be useful elsewhere. New code list is proposed, with terms like - river basin - lake - river

ATTRIBUTES
- catchment - canopy - epilimnion - hypolimnion - troposphere - tropopause - stratosphere - thermocline (useful as a general term, but needs "domain") - boundary layer (useful as a general term, but needs "domain") - ... [Is static True. Containment is Not Specified.]
🔑 featureName : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) free text field, e.g., "Danube Basin", "Amazon rain forest", "Amazon river Basin" [Is static True. Containment is Not Specified.]
ASSOCIATIONS
✎ Association (direction: Unspecified) Source: Public (Class) observation Target: Public (Class) domain

environment

Class in package 'WMDR2-core'

Optional facility-environment object containing temporal climate, surface-cover, population, and topography/bathymetry histories.

environment

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-02. Last modified 2026-06-02

ATTRIBUTES
🔑 temporalClimateZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Temporal climate-zone history. [Is static True. Containment is Not Specified.]
🔑 temporalPopulation : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]
🔑 temporalSurfaceCover : Private

ATTRIBUTES	
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
Temporal surface-cover history.	[Is static True. Containment is Not Specified.]
 temporalSurfaceRoughness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 topographyBathymetry : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public describes (Class) facility Cardinality: [1]	Target: Public is_described_by (Class) environment Cardinality: [0..1]

facility

Class in package 'WMDR2-core'

Facility properties payload of the WMDR2 full record. Root Feature geometry/time are modeled on fullRecord, not here.

facility

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES	
 contacts : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	
Facility-level contacts.	[Is static True. Containment is Not Specified.]
 created : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
Record creation date/time metadata.	[Is static True. Containment is Not Specified.]
 declaredReportingStatus : Private	[Is static True. Containment is Not Specified.]
 description : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	

ATTRIBUTES	
Facility description. [Is static True. Containment is Not Specified.]	
externalIds : Private Alias: 3-06 (primary & additional) WSIs Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) External identifiers such as WMO:WIGOS. [Is static True. Containment is Not Specified.]	
facilityType : Private Alias: 3-04 facilityType Compact WMDR facility type code value. [Is static True. Containment is Not Specified.]	
geometry : Private Alias: 3-07 current geolocation geometry = geolocation = coordinates + elevation [Is static True. Containment is Not Specified.]	
id : Private Alias: 3-06 (primary) WSI to prefixed with name space "facility:" [Is static True. Containment is Not Specified.]	
keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Discovery/search keywords retained in the core JSON. [Is static True. Containment is Not Specified.]	
links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Related links. [Is static True. Containment is Not Specified.]	
temporalTerritory : Private Alias: 3-02 temporal Territory Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) mandatory for fixed facilities, optional for mobile facilities [Is static True. Containment is Not Specified.]	
title : Private Alias: 3-03 facilityName Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) properties.title = 3-03 facilityName	

ATTRIBUTES	
	[Is static True. Containment is Not Specified.]
 type : Private Facility properties type; const facility.	[Is static True. Containment is Not Specified.]
 updated : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Record update date/time metadata.	[Is static True. Containment is Not Specified.]
 wmoRegion : Private Alias: 3-01 wmoRegion Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) mandatory for fixed facilities, optional for mobile facilities.	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified) 9-01 responsibleParty	
Source: Public supervises (Class) facility Cardinality: [0..*]	Target: Public is_supervised_by (Class) contact Cardinality: [1..*]
 Association (direction: Unspecified)	
Source: Public groups (Class) facility Cardinality: [0..*]	Target: Public belongs_to (Class) facilitySet Cardinality: [0..*]
 Association (direction: Unspecified)	
Source: Public is_hosted_by (Class) facility Cardinality: [1]	Target: Public hosts (Class) deployment Cardinality: [1..*]
 Association (direction: Unspecified) 3-07 temporalGeometry	
Source: Public describes (Class) facility Cardinality: [1]	Target: Public has_geolocation (Class) temporalGeometry Cardinality: [0..1]
 Association (direction: Unspecified)	
Source: Public is_hosted_by (Class) facility Cardinality: [1]	Target: Public hosts (Class) observation Cardinality: [1..*]
 Association (direction: Unspecified)	

ASSOCIATIONS	
Source: Public (Class) facility	Target: Public (Class) temporalProgramAffiliation
✍ Association (direction: Unspecified)	
Source: Public describes (Class) facility Cardinality: [1]	Target: Public is _described_by (Class) environment Cardinality: [0..1]

facilitySet

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

facilitySet

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES	
🔒 description : Private	[Is static True. Containment is Not Specified.]
🔒 id : Private	[Is static True. Containment is Not Specified.]
🔒 title : Private	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
✍ Association (direction: Unspecified)	
Source: Public groups (Class) facility Cardinality: [0..*]	Target: Public belongs_to (Class) facilitySet Cardinality: [0..*]

instrument

Class in package 'WMDR2-core'

Describes instrument capabilities, not a specific instrument instance.
For semantic consistency, terms like observedProperty and observedGeometry are used, even though they just describe the instrument capabilities, not the actual observations. With this, instrument becomes re-usable between deployments and even facilities.

instrument

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES

ATTRIBUTES	
5-09 manufacturer : Private	[Is static True. Containment is Not Specified.]
5-09 model : Private	[Is static True. Containment is Not Specified.]
id : Private	[Is static True. Containment is Not Specified.]
observedGeometry : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Specify using http://codes.wmo.int/wmdr/Geometry	[Is static True. Containment is Not Specified.]
observedProperty : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Specify as an array of values taken from http://codes.wmo.int/wmdr/ObservedVariable or as free text description.	[Is static True. Containment is Not Specified.]
verticalRange : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Specify using an array of [lower boundary, upper boundary] in meter. This is a capability of an instrument as specified, not the actually accessible vertical range.	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
Association (direction: Unspecified) 5-09 instrument Source: Public is used by (Class) deployment Cardinality: [1..*] Target: Public uses (Class) instrument Cardinality: [0..1]	

observation

Class in package 'WMDR2-core'

WMDR2 core JSON class: observation.

observation

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-06-02

ATTRIBUTES
contacts : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

ATTRIBUTES	
Observation-level contacts.	[Is static True. Containment is Not Specified.]
 deployments : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Deployment id references.	[Is static True. Containment is Not Specified.]
 id : Private Observation identifier, e.g. observation:<observedVariable>.	[Is static True. Containment is Not Specified.]
 keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation keywords.	[Is static True. Containment is Not Specified.]
 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation links.	[Is static True. Containment is Not Specified.]
 observedGeometry : Private cf. http://codes.wmo.int/wmdr/Geometry	[Is static True. Containment is Not Specified.]
 observedProperty : Private Alias: OMS:ObservedProperty, iop:Variable The observedVariable is constructed following I-ADOPT ontology approaches, with elements described in a unified vocabulary hosted at codes.wmo.int/wmdr2. NB: observedVariable will be a LONG list of entries, each record comprised of notation, name, description. The notation is a unique identifier, not necessarily human-readable. The name is potentially human-readable. The description is a fully fledged definition, human-readable and describing all relevant aspects of an observed variable. It is derived from the observedProperty, the object of interest, the matrix and domain, as well as constraints. Matrix and constraints are defined in controlled vocabularies that may be domain-specific. Examples include: - observableProperty: state variable (e.g., temperature, pressure, leaf-area-index (LAI), ...), or another property (e.g., thickness, concentration, scattering coefficient, color, ...) - objectOfInterest: sea ice, CO2, aerosol particles, chlorophyll - observablePropertyConstraints: a conventional level above ground (e.g., 2m above ground for temperature measurements), a size fraction (e.g., PM2.5), a specific wavelength (e.g., 635 nm), a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text. - matrix: atmosphere, interstitial air in the root space, air in an ice core, sea surface, soil, dry air, ... These elements will be supported by code lists from codes.wmo.int/wmdr2. Applications may want to guide users to select a specific observedVariable by offering groupings, filtering, or tree-like visualizations according to matrix, objectOfInterest, etc.	

ATTRIBUTES	
	[Is static True. Containment is Not Specified.]
 programAffiliation : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Program-affiliation timeline fragments inherited from WMDR1 observations.	[Is static True. Containment is Not Specified.]
 temporalDataPolicy : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Optional temporal data-policy history.	[Is static True. Containment is Not Specified.]
 time : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Observation temporal extent, date-resolution interval or null.	[Is static True. Containment is Not Specified.]
 title : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Human-readable title derived from domain, geometry, observed variable, and optional label.	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public (Class) observation	Target: Public (Class) domain
 Association (direction: Unspecified) 7-03 Temporal reporting period Source: Public describes (Class) observation Cardinality: [1]	Target: Public uses (Class) reporting Cardinality: [0..1]
 Association (direction: Unspecified) Source: Public describes (Class) observation Cardinality: [1..*]	Target: Public uses (Class) deployment Cardinality: [1..*]
 Association (direction: Unspecified) Source: Public is_hosted_by (Class) facility Cardinality: [1]	Target: Public hosts (Class) observation Cardinality: [1..*]

procedure

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

procedure

Version 1.0 Phase 1.0 Proposed

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ASSOCIATIONS	
Association (direction: Unspecified)	
Source: Public describes (Class) deployment Cardinality: [1..*]	Target: Public uses (Class) procedure Cardinality: [0..1] ObservingProcedure(s)

reporting

Class in package 'WMDR2-core'

Observation reporting metadata represented as aligned arrays. temporalTimeliness is itself a temporal wrapper.

reporting

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES	
 dataFormat : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of compact data-format values. [Is static True. Containment is Not Specified.]	
 dataPolicy : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of data-policy objects, preserving attribution when present. [Is static True. Containment is Not Specified.]	
 internationalExchange : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of international-exchange flags. [Is static True. Containment is Not Specified.]	
 levelOfData : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of compact level-of-data values. [Is static True. Containment is Not Specified.]	
 numberOfObservationsInReportingInterval : Private	

ATTRIBUTES	
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array copied from WMDR reporting metadata when present. [Is static True. Containment is Not Specified.]	
referenceDatum : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of reference-datum objects. [Is static True. Containment is Not Specified.]	
referenceTimeSource : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of reference-time-source values. [Is static True. Containment is Not Specified.]	
temporalAggregate : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of reporting/aggregation periods, renamed from temporalReportingInterval. [Is static True. Containment is Not Specified.]	
temporalTimeliness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Optional wrapper with timeliness durations and dates. [Is static True. Containment is Not Specified.]	
timeStampMeaning : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of timestamp-meaning values. [Is static True. Containment is Not Specified.]	
uom : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array of compact unit code values. [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
✍ Association (direction: Unspecified) 7-03 Temporal reporting period Source: Public describes (Class) observation Cardinality: [1] Target: Public uses (Class) reporting Cardinality: [0..1]	

schedule

Class in package 'WMDR2-core'

Reusable JSCalendar Event schedule used by deployment.temporalObservingSchedule references. WMDR-specific extensions are represented with wmo.int:* properties.

schedule
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES	
 @type : Private JSCalendar object type; const Event. [Is static True. Containment is Not Specified.]	
 duration : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Occurrence duration describing coverage window or sampling duration. [Is static True. Containment is Not Specified.]	
 recurrenceOverrides : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) JSCalendar recurrence override patches; null override values are invalid. [Is static True. Containment is Not Specified.]	
 recurrenceRules : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) JSCalendar recurrence rules. [Is static True. Containment is Not Specified.]	
 start : Private Canonical JSCalendar recurrence anchor; date 0001-01-01 with time from coverage start. [Is static True. Containment is Not Specified.]	
 timeZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) IANA time zone or UTC default. [Is static True. Containment is Not Specified.]	
 uid : Private JSCalendar-safe schedule uid. [Is static True. Containment is Not Specified.]	
 wmo.int:aggregation : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) WMDR-specific aggregation metadata. [Is static True. Containment is Not Specified.]	

ATTRIBUTES	
wmo.int:sampling : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) WMDR-specific sampling metadata preserved from dataGeneration.sampling. [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) Source: Public describes (Class) temporalObservingSchedule Cardinality: [1] Target: Public uses (Class) schedule Cardinality: [1..*]	

temporalGeometry

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalGeometry.

temporalGeometry

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES	
coordinates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Coordinate history array; each entry is lon, lat[, elevation]. [Is static True. Containment is Not Specified.]	
dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with coordinates. [Is static True. Containment is Not Specified.]	
methods : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) use http://codes.wmo.int/wmdr/GeopositioningMethod [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) 3-07 temporalGeometry Source: Public describes (Class) facility Cardinality: [1] Target: Public has_geolocation (Class) temporalGeometry Cardinality: [0..1]	

ASSOCIATIONS

temporalGeometry

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalGeometry.

temporalGeometry
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-06-12. Last modified 2026-06-02

ATTRIBUTES

coordinates : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Coordinate history array; each entry is lon, lat[, elevation].

[Is static True. Containment is Not Specified.]

dates : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Date array aligned with coordinates.

[Is static True. Containment is Not Specified.]

methods : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

use <http://codes.wmo.int/wmdr/GeopositioningMethod>

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified) 3-07 temporalGeometry

Source: Public (Class) deployment
Cardinality: [1]

Target: Public (Class) temporalGeometry
Cardinality: [0..1]

describes

has_geolocatino

temporalObservingSchedule

Class in package 'WMDR2-core'

When are observations made? Describes a temporal schedule/coverage, e.g. "from Feb-Oct on working days from 06-18 UTC".

Objects like

{"date": date, "schedule": str}, whereby "schedule" is a reference to a re-usable schedule object.

temporalObservingSchedule

ATTRIBUTES	
dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with observingSchedule. [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) Source: Public describes (Class) temporalObservingSchedule Cardinality: [1] Target: Public uses (Class) schedule Cardinality: [1..*]	
Association (direction: Unspecified) Source: Public describes (Class) deployment Cardinality: [1] Target: Public reports_with (Class) temporalObservingSchedule Cardinality: [0..*]	

temporalProgramAffiliation

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalProgramAffiliation.

temporalProgramAffiliation

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-02

ATTRIBUTES	
dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with programAffiliation. [Is static True. Containment is Not Specified.]	
programAffiliation : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Compact program-affiliation values. [Is static True. Containment is Not Specified.]	
reportingStatus : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Optional compact reporting-status values aligned with programAffiliation. [Is static True. Containment is Not Specified.]	

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public (Class) facility	Target: Public (Class) temporalProgramAffiliation